

Reduction of organic compounds

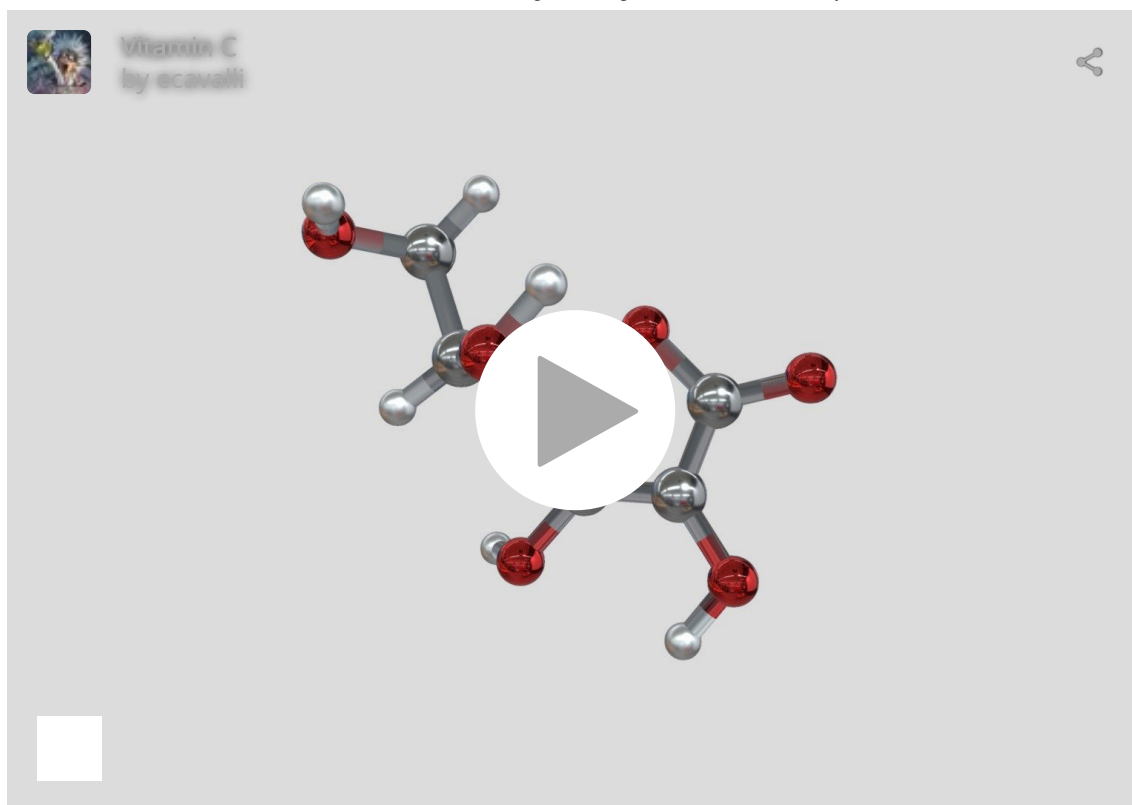
Learning outcomes

By the end of this section you should be able to deduce equations to show the reduction of carboxylic acids to primary alcohols via the aldehyde, and the reduction of ketones to secondary alcohols.

Vitamin C, also known as ascorbic acid, is a water-soluble vitamin that plays a crucial role in various biological functions. It is an essential nutrient, which means that it cannot be synthesised by the human body and must be obtained from the diet. Vitamin C is found in many fruits and vegetables, including citrus fruits, strawberries, kiwi, broccoli, and peppers.

Vitamin C is a six-carbon compound with the chemical formula $C_6H_8O_6$, belonging to the family of organic compounds known as polyols or sugar alcohols. Ascorbic acid contains both hydroxyl ($-OH$) and carbonyl ($C=O$) functional groups as shown in **Interactive 1**, which give it its unique chemical and biological properties.

It can be produced industrially by the reduction of carboxylic acids.



Interactive 1. Ascorbic acid, also known as vitamin C.

 More information for interactive 1

An interactive simulation features a three-dimensional ball-and-stick model of ascorbic acid. Users can rotate and manipulate this model in any direction. The carbon, oxygen, and hydrogen atoms are represented by spheres, and the connecting bonds are represented by thin rods.

The ascorbic acid structure features a pentagonal structure with an oxygen atom at the top, a carbon at the top right, a carbon at the bottom right, a carbon at the bottom left, and a CH group at the top left. All are connected by single bonds, except the carbon atoms on the bottom, which are connected to each other by a double bond.

The carbon atoms at the bottom are each connected to an OH group, and the carbon at the top right is double-bonded to an oxygen atom. The CH group on the top left is connected to another CH group, which is further bonded to an OH group and a CH₂ group. This CH₂ is bonded to an OH group.

This model helps users understand the structure, bonding, and geometry of ascorbic acid.

In the early 20th century, it was discovered that vitamin C deficiency was the cause of scurvy, and that scurvy could be prevented by consuming foods containing vitamin C. This led to the development of vitamin C supplements, which have been widely used since then as a dietary supplement. The industrial production of vitamin C is a complex process, but it highlights the conversion of carboxylic acids into aldehydes, ketones or alcohols through reduction reactions.

How is the reduction of carboxylic acids used to produce aldehydes, ketones or alcohols?

Reduction of organic compounds

As mentioned, carboxylic acids can be reduced back to aldehyde and primary alcohol in the presence of a reducing agent.

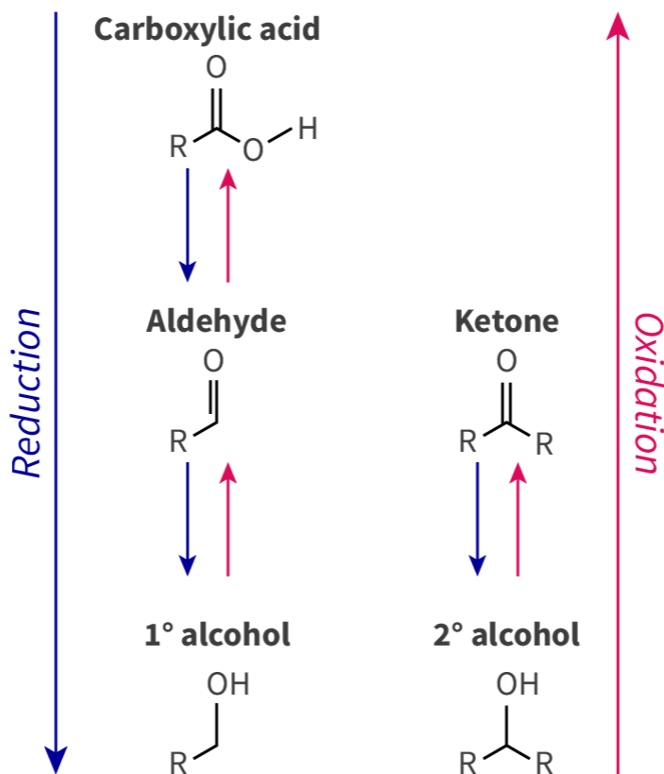


Figure 1. Reaction pathways between carboxylic acids, aldehydes, alcohols and ketones.

 More information for figure 1

The diagram presents the reaction pathways between carboxylic acids, aldehydes, alcohols, and ketones, illustrating flows of oxidation and reduction.

On the left, labeled 'Reduction,' the diagram begins with carboxylic acid, represented by a structural formula ($\text{R}-\text{C}(\text{O})\text{OH}$). Below it is 'Aldehyde' ($\text{R}-\text{C}(\text{O})\text{H}$) with an arrow pointing downward, indicating reduction. Further down is ' 1° alcohol' ($\text{R}-\text{CH}_2\text{OH}$), also connected via a downward arrow.

On the right, labeled 'Oxidation,' the pathway illustrates 'Ketone' ($\text{R}_2\text{C}=\text{O}$) converting back to ' 2° alcohol' (R_2CHOH), both connected with arrows indicating the direction of reactions.

Overall, the diagram emphasizes the reversible nature of these chemical transformations via redox processes, underscoring how carboxylic acids can be reduced to aldehydes and then to primary alcohols, while ketones can be oxidized and reduced to secondary alcohols.

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Recalling that reduction can involve the removal of oxygen or the addition of hydrogen, what element must the reducing agent have in order to perform its function?

The reducing agent needs to have hydrogen atoms to donate to the carboxylic acid, in order for reduction to occur. That can be in the form of hydride ions (H^-) or hydrogen ions (H^+).

Theory of Knowledge

The commercial synthesis of natural substances, such as vitamins, are chemically identical to those synthesized by plants. To what extent do chemists rely on nature for knowledge?

These compounds can be reduced back to the respective alcohol in reduction reactions using a suitable reducing agent. The partial positive charge on the carbon atom makes this group vulnerable to attack from a nucleophile, which is the hydride ion (H^-). The sources of the hydride ions are the reducing agents sodium borohydride (NaBH_4) and lithium aluminium hydride (LiAlH_4).

Study skills

Remember that names and formulas of specific reducing agents and the mechanisms will not be assessed.

Figure 2 shows the reduction of an aldehyde, propanal, to a primary alcohol, propan-1-ol. The reducing agent is sodium borohydride (lithium aluminium hydride can also be used) followed by the addition of an acidic solution (H^+).

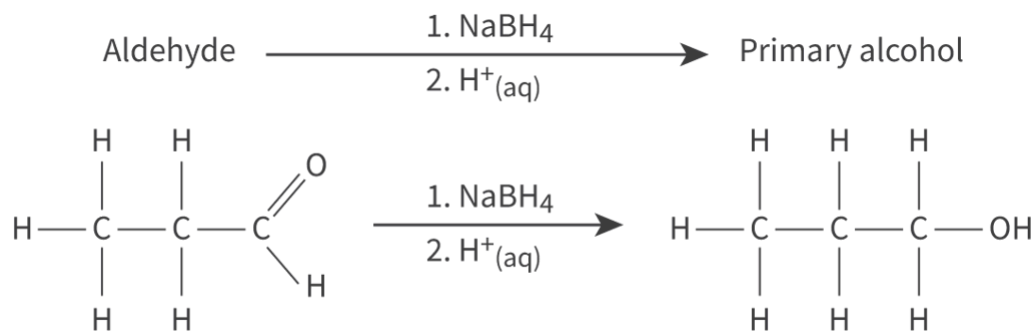


Figure 2. The reduction of the aldehyde propanal to the primary alcohol propan-1-ol.

 More information for figure 2

The image is a chemical reaction diagram illustrating the reduction of the aldehyde propanal to the primary alcohol, propan-1-ol. On the left, the structure of propanal is shown with a carbon chain: three carbon atoms with hydrogen atoms attached, and a double bonded oxygen atom at the end, indicating the aldehyde group. An arrow points to the right, labeled '1. NaBH₄, 2. H⁺(aq)', representing the reduction process. On the right, the structure of propan-1-ol is displayed, where the double-bonded oxygen has been replaced by an OH group, indicating the transformation into a primary alcohol.

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Figure 3 shows the reduction of a ketone, propanone, to a secondary alcohol, propan-2-ol. The reducing agent is sodium borohydride followed by the addition of an acidic solution. As with the reduction of aldehydes, it is also possible to use lithium aluminium hydride as the reducing agent.

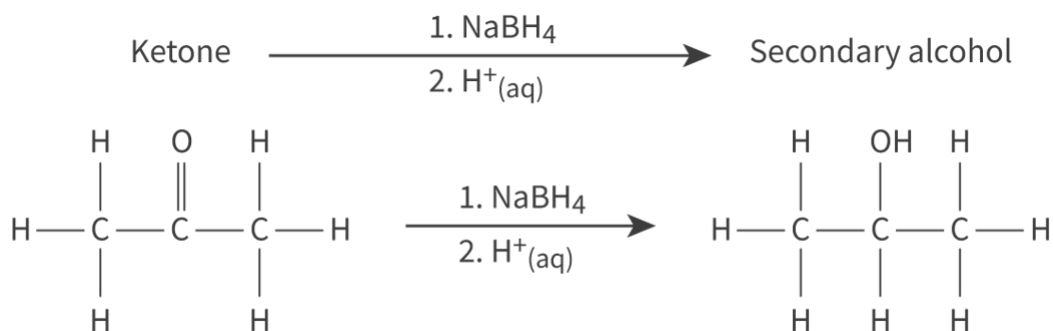


Figure 3. The reduction of the ketone propanone to the secondary alcohol propan-2-ol.

 More information for figure 3

The image is a chemical diagram depicting the reduction of propanone (a ketone) to propan-2-ol (a secondary alcohol). The diagram contains two chemical structures and a reaction arrow. On the left, the structure of propanone is shown, with a central carbon double-bonded to an oxygen and single-bonded to two methyl groups, forming the ketone group. An arrow labeled "1. NaBH₄ 2. H⁺(aq)" indicates the conversion process using sodium borohydride followed by an acid solution, leading to the structure on the right. The right structure represents propan-2-ol, where the central carbon is single-bonded to an -OH group and two methyl groups, indicating the transformation to an alcohol.

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Figure 4 shows the reduction of propanoic acid to the primary alcohol propan-1-ol.

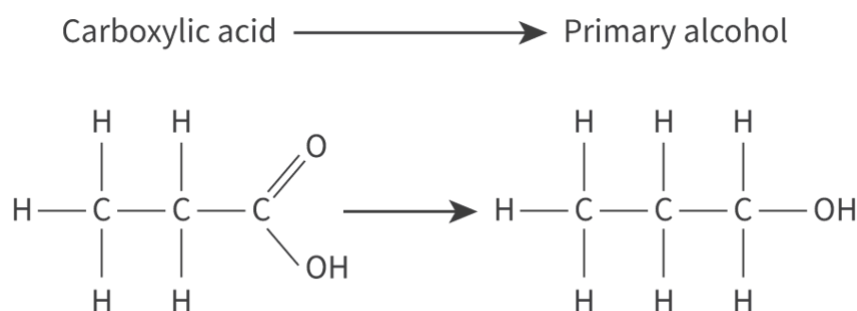


Figure 4. The reduction of propanoic acid to the primary alcohol propan-1-ol.

 More information for figure 4

The image is a chemical diagram depicting the reduction reaction of propanoic acid to propan-1-ol. On the left, there's a structural formula for propanoic acid, which includes three carbon atoms. The first two carbon atoms are each bonded to three hydrogen atoms. The third carbon atom is double-bonded to an oxygen atom and single-bonded to a hydroxyl group (-OH), indicating it is a carboxylic acid. An arrow points to the right, labeled from carboxylic acid to primary alcohol, indicating the reduction process.

On the right side, there's the structural formula for the primary alcohol propan-1-ol. It consists of three carbon atoms, with the first two carbon atoms each bonded to three hydrogen atoms. The third carbon atom is bonded to two hydrogen atoms and one hydroxyl group (-OH), showing that the acid has been reduced to an alcohol with the loss of one oxygen atom and gain of two hydrogen atoms.

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Worked example 1

Deduce the final product in the reduction of methanal.

Methanal is an aldehyde and therefore it will produce a primary alcohol through reduction so the final product will be methanol.

Worked example 2

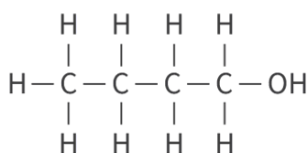
Deduce the final product in the reduction of hexanoic acid.

Hexanoic acid is a carboxylic acid and therefore it will produce a primary alcohol through reduction so the final product will be hexan-1-ol.

Worked example 3

Draw and name the final product in the reduction of butanal.

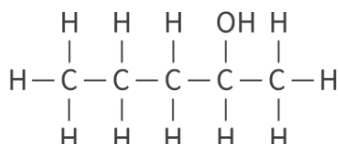
Butanal is an aldehyde and therefore it will produce a primary alcohol through reduction so the final product will be butan-1-ol.



Worked example 4

Draw and name the final product in the reduction of pentan-2-one.

Pentan-2-one is a ketone and therefore it will produce a secondary alcohol through reduction so the final product will be pentan-2-ol.



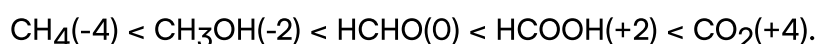
Making connections

How can oxidation states be used to show that the following molecules are given in increasing order of oxidation: CH_4 , CH_3OH , HCHO , HCOOH , CO_2 ? (see subtopic S3.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>)).

Oxidation state is a measure of the degree of oxidation of an atom in a molecule. Knowing that each molecule has an overall oxidation state of zero, we can assign the oxidation states of each atom in the given molecules:

- CH_4 : Carbon has an oxidation state of -4 as each hydrogen has an oxidation state of +1.
- CH_3OH : Carbon has an oxidation state of -2, as the oxygen has an oxidation state of -2, and each hydrogen has an oxidation state of +1.
- HCHO : Carbon has an oxidation state of 0, as the oxygen has an oxidation state of -2, and each hydrogen has an oxidation state of +1.
- HCOOH : Carbon has an oxidation state of +2, as each oxygen has an oxidation state of -2, and each hydrogen has an oxidation state of +1.
- CO_2 : Carbon has an oxidation state of +4 and each oxygen has an oxidation state of -2.

From these oxidation state assignments, we can see that the given molecules are ordered from increasing oxidation state as:



Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair/group activity

Research examples of pharmaceuticals that are synthesised using reduction reactions, such as ibuprofen and naproxen. Research about the synthesis of these drugs and focus on the role of reduction reactions in their synthesis.

Share your findings with a partner and reflect on what you have learned.